Zoom In, Scale Out

Metric depth model gulo ki digital zoom er physics mane?

Research overview · Rabeya Noor · target CVPR 2027 · 20 October 2026

Ek line e: Chhobi crop kore boro korle (digital zoom) focal length thik s gun bare, kintu kono jinish kachhe ashe na. Amra dekhiyechi beshirbhag metric depth model zoom korle depth kom bole: shobcheye popular Depth Anything V2 3x zoom e depth prai 1/3 bole (α = 0.95). Ete label lage na, theory ache, ar label chhara fine-tuning (EFT+B) diye α prai 0 te ana jay, kintu shadharon chhobite accuracy kichuta kome.

1. Shomossha

Self-driving, robot, AR ar phone camera te ekta chhobi theke protita jinish koto **meter** dure, sheta bolte metric depth model lage (Depth Anything, Depth Pro, MoGe, UniDepth, Metric3D, Depth Anything 3). Meter e bolte hole camera er focal length jana dorkar. Karon wide lens e kachher jinish ar telephoto lens e durer jinish ekdom ek rokom dekhate pare. Model gulo camera chhara shudhu pixel dekhe eta anuman kore. Proshno holo: ora ki asholei camera bujhe, naki shudhu "boro dekhale kachhe" shortcut use kore?

2. Mul idea: digital zoom ekta free, shothik ruler

Chhobir ekta box crop kore abar ager size e resize korle (digital zoom), chhobi ta ekdom shothik bhabe ekta notun pinhole camera theke tola chhobir moto hoy: f' = s·f, principal point shore jay, kintu **depth bodlay na**. Tai label chhara test kora jay: zoom er age ar pore model nijer depth ar focal er sathe tulona korlei hoy.

Shonkhya	Mane	Thik model	Shortcut model
α (depth exponent)	D(zoom) / D(original) ≈ s^−α	0	1 (2x zoom e depth ordhek)
β (focal exponent)	f(zoom) / f(original) ≈ s^β	1	0 (focal bodlay na)

3. Theory

3.1 Tinta invariant

- α = 0 → depth thik thake;
- β = 1 → 3D shape thik thake;
- α + β = 1 → shudhu pasher (lateral) map thik thake, depth ba shape na. "Boro dekhale kachhe" reasoning thik ei line e pore: Ḋ = ḟ·L/ℓ hole zoom e Ḋ ∝ s^(β−1).

3.2 Zoom–squash ambiguity (notun)

Focal s·f e tola chhobi pixel-by-pixel sheyi scene er chhobi, jeta depth er dik borabor s gun chepta kora: u = s·f·X/Z = f·X/(Z/s). Chepta scene e jinish er width ek thake, shudhu gobhirota bhul hoy (4.5 m er gari 2x e 2.25 m lomba hoye jay). Tai shudhu size cue diye zoom dhora oshombhob, 3D shape prior lage. Ar shoja dewal (fronto-parallel plane) e kono model-i camera na jene zoom dhorte parbe na.

3.3 Stale K

Metric3D er moto camera-input model e zoom er por purono K dile network ekoi pixel dekhe, shudhu shesh e scale bodlay. Tai α(stale) = α(true) + 1, eta ganit diye proman kora jay. Data te median tofat 1.000.

4. Experiment setup

Ki	Details
Model (main)	DA-V2-S/B, ZoeDepth (pixels only) · Depth Pro, MoGe-2-S, UniDepthV2-S (predicts camera) · Metric3D v2-S, thik K ar stale K (takes camera)
Model ladder	DA-V2-L, MoGe-2-B/L, UniDepthV2-B/L, UniK3D-L, Metric3D v2-L, Depth Anything 3 (Nested-Giant, Metric-L, L), VGGT-1B
Data	KITTI val selection (LiDAR), NYUv2 test (Kinect), DIODE val indoor + outdoor (laser). Protita theke 200 chhobi, shob model e ekoi chhobi (800 chhobi/model)

Ki	Details
Zoom	1, 1.25, 1.5, 2, 3x majhkhane · off-centre: 4 kona, 50 chhobi (KITTI, NYU)
Control	blur control (zoom er moto jhapsa, field of view ek) · geometry check (pixel error 0) · unzoomed prediction ekoi pixel e (fair reference) · optical vs digital zoom, bilinear, JPEG · result verify tool (140 file, shob check pass)
Statistics	95% bootstrap CI (paper e scene-level cluster bootstrap) · paired Wilcoxon signed-rank + Holm correction · Spearman with bootstrap CI
Compute	Kaggle T4 GPU; laptop e shudhu analysis

5. Result 1: zoom audit

Model	Group	α [95% CI]	β	α+β	Blur control	AbsRel 1x	AbsRel 3x
DA-V2-S	pixels only	0.95 [0.93, 0.99]	–	–	1.3%	0.317	0.652
DA-V2-B	pixels only	0.85 [0.82, 0.87]	–	–	0.8%	0.313	0.599
ZoeDepth	pixels only	0.61 [0.58, 0.64]	–	–	0.7%	0.310	0.539
Depth Pro	predicts camera	0.31 [0.27, 0.34]	0.65	0.97	2.8%	0.310	0.459
MoGe-2-S	predicts camera	0.36 [0.34, 0.38]	0.56	0.94	3.1%	0.274	0.465
UniDepthV2-S	predicts camera	0.34 [0.30, 0.40]	0.45	0.87	4.0%	0.824	0.792
Metric3D v2-S	takes camera (true K)	-0.07 [-0.08, -0.06]	–	–	1.6%	0.142	0.358
Metric3D v2-S (stale K)	takes camera (stale K)	0.93 [0.92, 0.94]	–	–	1.6%	0.142	0.628

Note: ei table er AbsRel 4 ta dataset er gor (DIODE-outdoor er boro error shoho). Section 7 er EFT table e protita dataset alada, ar sheta 392 px input e chale. Tai DA-V2-S er 1x AbsRel ekhane 0.317, ar EFT table e KITTI te 0.138. KITTI te default preprocessing e eta 0.133, mane duita mile jay. CI ekhane image-level; paper e scene-level cluster bootstrap use kora hoy, jeta aro chouro.

Ja pelam:

- Pixels-only model gulo prai puro shortcut use kore: DA-V2-S α = 0.95.
- Camera predict kora model gulo ordhek bujhe: α ≈ 0.31–0.36, ar focal 3x er jaygay 1.63–2.04x bole.
- Depth Pro e protita chhobite α+β er median 0.97, mane theory er "lateral invariant" line e prai hubohu thake.
- Thik K pele Metric3D prai equivariant (α = -0.07), stale K dile pixels-only model er moto (0.93).
- Blur control e depth bodlay mattro 0.7–4%, tai dosh interpolation er na, camera bojhar.

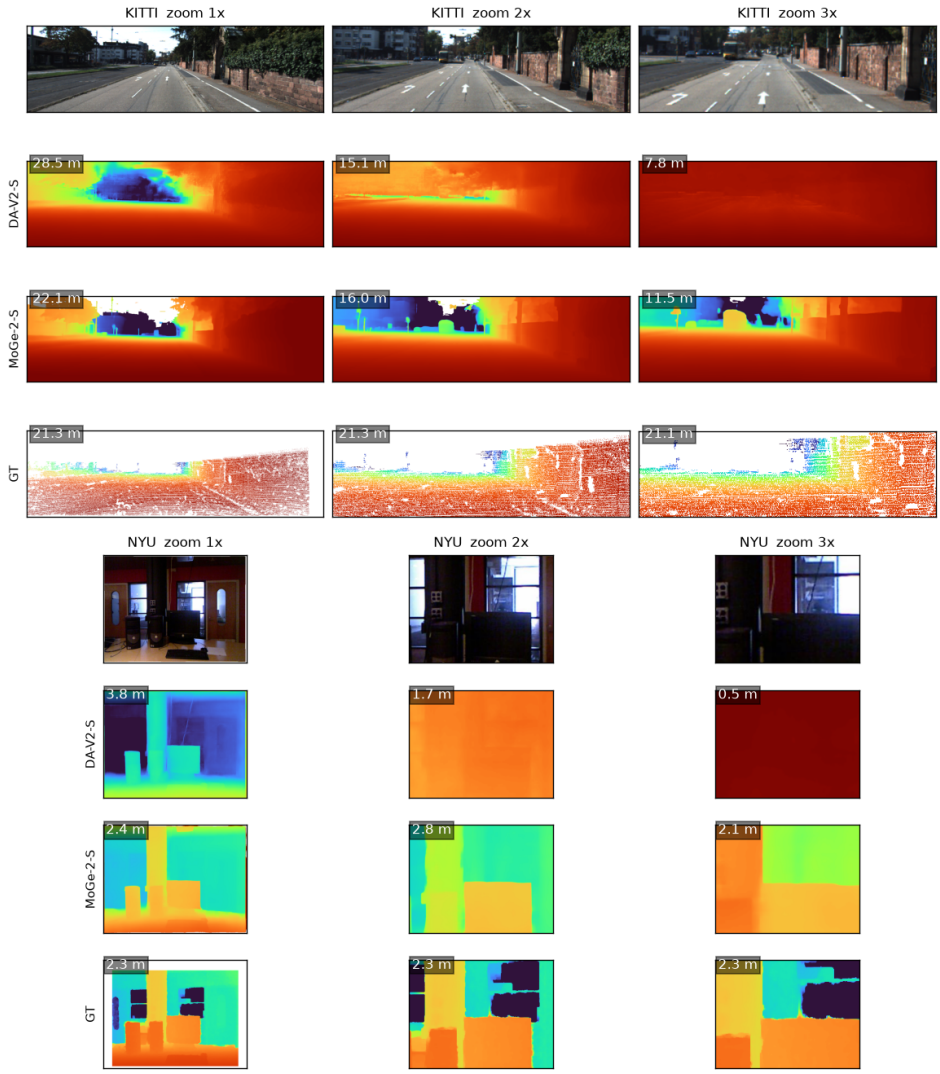


Figure 1. Ekoi scene 1x, 2x, 3x zoom e. Ek row er shob depth panel ekoi rong er scale e (meter). DA-V2-S zoom korle "kachhe" chole jay (KITTI 28.5 → 7.8 m, GT 21 m); MoGe-2 ordhek.

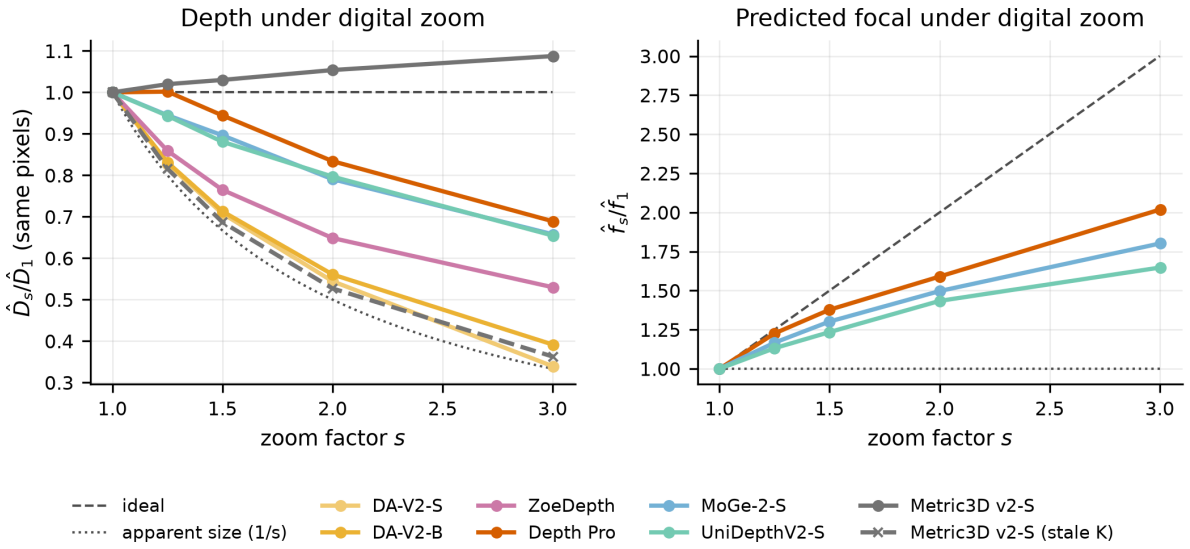


Figure 2. Zoom e depth (bame) ar predicted focal (dane), 800 chhobir median.

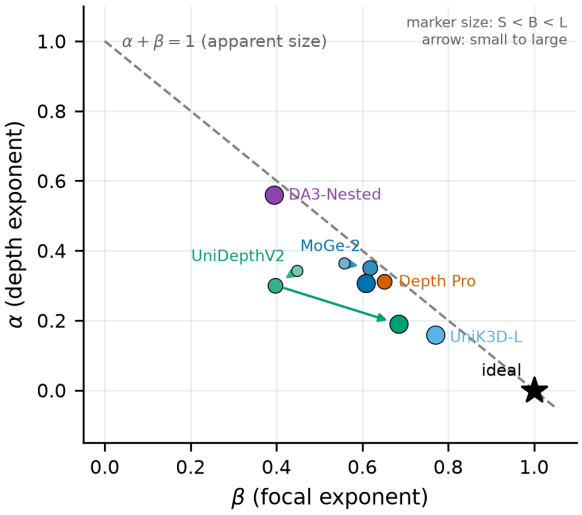


Figure 3. α–β plane. Dashed line: α+β = 1 (apparent size reasoning). Tara: ideal.

6. Result 2: label chhara score diye error anuman

|α| (label lage na) ar 3x zoom e AbsRel er Spearman ρ = 0.55 [0.18, 0.86], 32 ta model×dataset jora. Dataset er bhitore KITTI 0.98, NYU 0.95, DIODE-in 0.88, kintu DIODE-out -0.38. Shadharon (1x) accuracy predict kore na (ρ = -0.09). **Shotti kotha:** zoom e depth shrink korle error bare-i, tai sambandha kichuta nijer theke ashe. Claim ta tai choto: "zoom e kon model koto bhul korbe, label chhara bola jay."

7. Result 3: label chhara fine-tuning (EFT ar EFT+B)

Frozen copy (teacher) zoom chhara chhobi te ja bole, student ke zoom kora chhobi te ta-i bolte shekhai. Kono depth label ba camera info lage na. Training data: 7 ta KITTI raw drive (test drive er sathe overlap nai, code nijei check kore) ar NYUv2 er 795 ta train chhobi. DIODE kokhono training e dekhano hoy na.

Notun finding, resolution shortcut: shudhu zoom loss diye train korle (EFT) model shikhe ney "jhapsa chhobi = zoom". Blur control e KITTI te depth 30% bodle jay. EFT+B te ekta blur-invariance term jog kori: jhapsa (kintu zoom na) chhobite ager depth-i bolte hobe. Eta Zhong et al. (ICCV 2023) e nai.

Dataset	Model	α	AbsRel 1x	AbsRel 3x	Blur shortcut
KITTI	DA-V2-S (frozen)	1.03	0.138	0.665	1.1%
KITTI	blur-only FT	0.82	0.160	0.538	0.2%
KITTI	Zhong et al.-style	0.18	0.222	0.143	25.8%
KITTI	EFT	0.18	0.156	0.158	29.6%
KITTI	EFT+B (ours)	0.05	0.175	0.178	4.9%
KITTI	EFT+B, λ=3	0.07	0.173	0.162	6.7%
NYUv2	DA-V2-S (frozen)	0.98	0.344	0.563	0.7%
NYUv2	blur-only FT	0.56	0.384	0.298	0.2%
NYUv2	Zhong et al.-style	0.11	0.453	0.314	4.1%
NYUv2	EFT	0.12	0.313	0.226	4.2%
NYUv2	EFT+B (ours)	0.07	0.301	0.266	0.4%
NYUv2	EFT+B, λ=3	0.05	0.360	0.293	0.3%
DIODE-in (unseen)	DA-V2-S (frozen)	0.70	0.280	0.558	1.0%
DIODE-in (unseen)	blur-only FT	0.44	0.404	0.385	0.3%
DIODE-in (unseen)	Zhong et al.-style	0.39	0.419	0.358	0.6%
DIODE-in (unseen)	EFT	0.39	0.368	0.364	0.6%
DIODE-in (unseen)	EFT+B (ours)	0.37	0.400	0.367	0.4%
DIODE-in (unseen)	EFT+B, λ=3	0.35	0.432	0.365	0.4%

Dataset	Model	α	AbsRel 1x	AbsRel 3x	Blur shortcut
DIODE-out (unseen)	DA-V2-S (frozen)	0.65	0.731	0.551	0.6%
DIODE-out (unseen)	blur-only FT	0.45	0.802	0.622	0.2%
DIODE-out (unseen)	Zhong et al.-style	0.37	0.926	0.682	1.6%
DIODE-out (unseen)	EFT	0.39	0.877	0.648	1.5%
DIODE-out (unseen)	EFT+B (ours)	0.40	1.000	0.676	0.4%
DIODE-out (unseen)	EFT+B, λ=3	0.41	0.971	0.647	0.4%

- EFT+B prai puro equivariant (KITTI te α 0.05, NYU te 0.07), ar blur shortcut prai chole geche.
- Trade-off: KITTI te 1x AbsRel 0.138 → 0.175, DIODE e aro beshi (training e dekhe nai); NYU te bhalo hoy.
- λ = 3 (3 gun shokto anchor) eta thik kore na: KITTI 0.173, NYU kharap hoy (0.360). Trade-off ta weight tune kore shoray jay na.
- Blur-only (zoom term chhara) α beshi komay na (KITTI 0.82): jhapsa chhobi dekhano camera shekhay na.
- Zhong et al.-style (frozen teacher er bodole student nijer prediction target) α komay (0.18), kintu blur shortcut fire ashe (26%) ar 1x AbsRel shobcheye kharap (0.222). Frozen teacher, zoom term ar blur term, tinta-i lage.
- "Anchor-only" baseline asole frozen model (student = teacher, tai gradient 0), tai blur-only control.

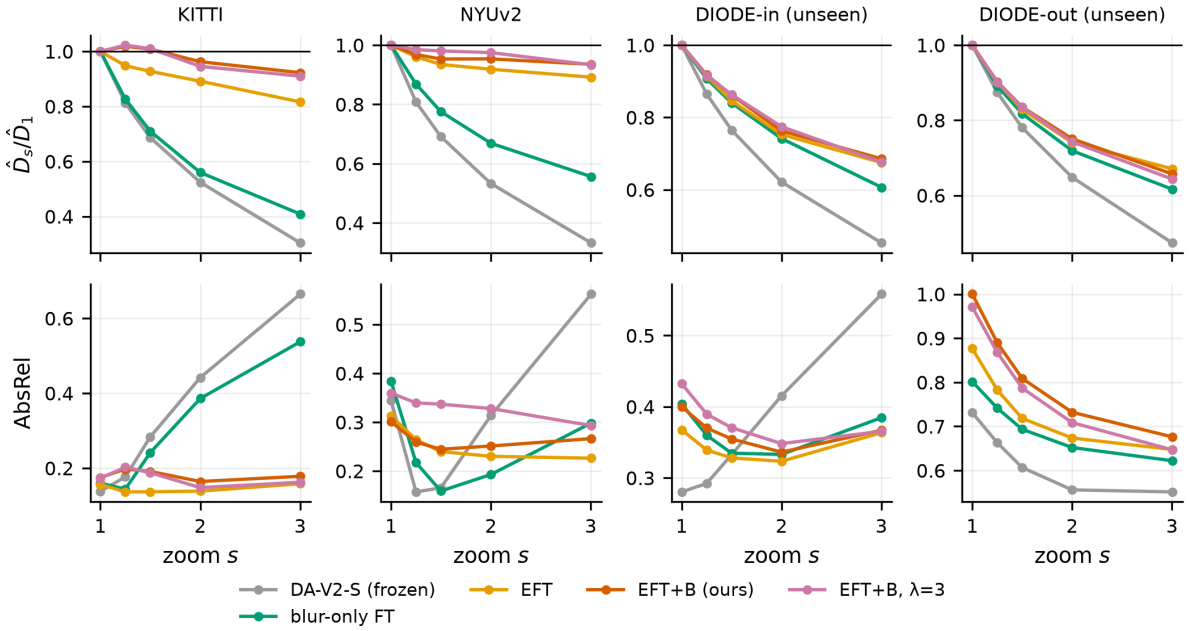


Figure 5. Upore: zoom e depth ratio (1 = ideal). Niche: AbsRel. Dhushor: frozen, holud: EFT, lal: EFT+B, golapi: EFT+B λ = 3, sobuj: blur-only. Note: alada zoom e AbsRel alada pixel e mapa; NYU te frozen model 1.25–1.5x e 1x er cheye bhalo AbsRel pay, karon 1x e depth beshi bole ar zoom shrink kakatale thik er kachhe ane. Tai AbsRel na, α dekhte hobe.

8. Model boro ba notun hole ki thik hoy? (model ladder)

Model	Params	α	β	chhobi
DA-V2-S	25M	0.95	–	800
DA-V2-B	97M	0.85	–	800
DA-V2-L	335M	0.73	–	400
MoGe-2-S	35M	0.36	0.56	800
MoGe-2-B	105M	0.35	0.62	400
MoGe-2-L	331M	0.31	0.61	400
UniDepthV2-S	34M	0.34	0.45	800
UniDepthV2-B	114M	0.30	0.40	400
UniDepthV2-L	354M	0.19	0.69	400
UniK3D-L	359M	0.16	0.77	400

Model	Params	α	β	chhobi
Depth Pro	952M	0.31	0.65	800
DA3-Nested	1,690M	0.56	0.39	200
DA3-L (relative)	411M	–	0.37	200
VGGT-1B (relative)	1,257M	–	0.81	200
DA3-Metric-L	334M	-0.05	–	200
Metric3D v2-S	37M	-0.07	–	800
Metric3D v2-L	412M	-0.03	–	200

Protita model nijer sample e (main model 200, boro variant 100, DA3/VGGT/Metric3D-L 50 chhobi prati dataset). Paper e ekoi chhobi te paired tulona-o ache (151 ta chhobi), result ek.

- DA-V2 boro hole α ektu kome (0.95 → 0.73), kintu L model-o 3x e ordheker kom depth bole.
- MoGe-2 er size bariye labh nai (0.36, 0.35, 0.31); UniDepthV2 L e bhalo (0.19); UniK3D-L shobcheye kache (α 0.16, β 0.77).
- Shobcheye boro DA3-Nested (1,690M) α = 0.56, chhoto camera model gulor cheye kharap, ZoeDepth er kachhe.
- Shothik camera dile DA3-Metric-L (-0.05) ar Metric3D-L (-0.03) depth thik rakhe.
- **Mane:** model koto boro ta na, camera model e dhoke kina setai ashol.

9. Control, principal point ar ZoomFix

Model	Optical 2x	Digital 2x	Bilinear 2x	JPEG
DA-V2-S	0.58	0.55	0.54	1.00
Depth Pro	0.83	0.75	0.82	1.00
MoGe-2-S	0.77	0.82	0.79	1.05
Metric3D v2-S	1.05	1.04	1.05	0.99

Depth ratio (1 = bodlay na). Real detail shoho optical zoom-o DA-V2-S er depth prai digital zoom er motoi komay. Tai dosh detail haranor na, camera bojhar. Shudhu Depth Pro te detail kichuta matter kore.

Principal point: 2x quadrant crop e principal point image width er ordhek shore jay. MoGe-2 er error 0.496, UniDepthV2 0.494, "shob shomoy centre" bolle 0.496. Duita model-i shore jawa dhore na.

Model	Dataset	ê (label chhara)	3x as is	3x + ZoomFix	Target (1x, ekoi pixel)
DA-V2-S	DIODE-in	0.75	0.598	0.365	0.263
DA-V2-S	DIODE-out	0.69	0.557	0.661	0.605
DA-V2-S	KITTI	1.25	0.758	0.384	0.149
DA-V2-S	NYUv2	1.12	0.675	0.495	0.205
Depth Pro	DIODE-in	0.19	0.399	0.421	0.201
Depth Pro	DIODE-out	0.12	0.635	0.638	0.665
Depth Pro	KITTI	0.31	0.258	0.278	0.160
Depth Pro	NYUv2	0.44	0.449	0.223	0.121
MoGe-2-S	DIODE-in	0.44	0.493	0.364	0.257
MoGe-2-S	DIODE-out	0.38	3.037	4.396	0.379
MoGe-2-S	KITTI	0.44	0.536	0.262	0.255
MoGe-2-S	NYUv2	0.12	0.272	0.227	0.129

ZoomFix: zoom factor jana thakle (phone EXIF) depth ke s^ê diye gun kori; ê label chhara ordhek chhobi te mapa, baki ordheke score. DA-V2-S e KITTI, NYU, DIODE-in e zoom er error er 1/3 theke 2/3 shore jay. DIODE-out e mean bare, karon kichu chhobite agei boro error (heavy tail), scale up korle ta aro bare. Eta ekta stop-gap, camera-aware model er bikolpo na.

10. Shotti negative result

- Theory er "soft" prediction (scene e depth beshi chhoriye thakle α kombe) data te pawa jay nai (camera model e ρ ≈ 0).

- Label-free score shadharon accuracy predict kore na, shudhu zoom e accuracy.
- UniDepthV2-S DIODE-outdoor e lomba error tail dey (prai 10% chhobite 6x beshi depth); α er upor eta prabhab felbe na.
- Stale-K result notun na (Kim & Lee 2026 er kaj ke support kore).

11. Agerkar kaj ar amader notun ongsho

Paper	Ki koreche	Amader sathe tofat
Zhong et al., ICCV 2023	crop-resize equivariance, self-consistency loss	camera/metric nai; amra α, β, theory, notun model, shortcut + EFT+B
Kim & Lee, Electronics 2026	stale intrinsics e camera-input model er bhul	amader stale-K tader support kore
Roll-robust depth (2026)	camera roll	roll e intrinsics bodlay na, zoom e bodlay
KPE (2023), PlaneDepth (CVPR 2023)	crop er perspective ambiguity	test/training input; zoom audit na
FS-Depth (2023)	crop diye focal augment, label shoho	amader label ar focal input chhara
DepthCues (CVPR 2025)	depth cue probing (size shoho)	amra size cue er physical meaning (α)

Amader notun ongsho: (1) metric depth er zoom audit with α ar β; (2) α+β = 1 line ar zoom–squash theory; (3) teen dhoroner model (pixels only / predicts camera / takes camera) ek scale e; (4) label-free proxy; (5) resolution shortcut finding ar EFT+B; (6) principal point test ar ZoomFix (EXIF zoom factor diye training chhara fix); (7) model ladder (DA3, VGGT, UniK3D shoho).

12. Code

- GitHub (private): github.com/rabeyanoor/zoom-equivariance
- 28 ta unit test (zoom geometry, audit toy model, analysis, ZoomFix), ruff lint, GitHub Actions CI
- Vectorised analysis (loop er cheye 150 gun druto, test e hubohu ek)
- verify_results.py: protita CSV te shob zoom, duplicate nai, shob model ekoi chhobi
- Paper er shob number make_paper_numbers.py diye CSV theke, haate kichu lekha hoy na
- Protita Kaggle run er log logs/kaggle/ e

13. Baki kaj

Kaj	Obostha
Shob Kaggle run (main audit, ladder c1 + c2, controls, ZoomFix, EFT ablation)	shesh
Paper (8 page + references) ar supplement, shob number script theke	shesh, draft
Lekha ta abar pora, choto kora, figure polish	baki
(Optional) phone e 1x/2x/5x zoom chhobi, tape diye mapa dure	tumi korbe
Overleaf e PDF, supervisor review, CVPR submit (~7–16 Nov)	shesh e